

Thuc Cao

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EDUCATION

Rutgers University – New Brunswick, NJ

Aug 2024 – May 2026

B.S. in Data Science, Minor in Economics & Statistics

GPA: 3.29

- Relevant Coursework: Regression Methods, Economic Forecasting and Big Data, Time Series and Financial Econometrics, Theory of Probability, Theory of Statistics, Game Theory, Stochastic Process, Combinatorics.

Edmonds College - Lynnwood, WA

Aug 2021 – June 2024

A.S. in Computer Science

GPA: 3.58

- Relevant Coursework: Python, Java, C++, MySQL, Calculus I-III, Linear Algebra, Differential Equation.

EXPERIENCE

Data Scientist Intern

June 2024 – August 2024

Hitachi Digital Services

Ho Chi Minh City, Vietnam

- Improved **AI chatbot task-resolution accuracy**, measured by a **10% accuracy increase**, by developing **intent classification** and **workflow-based response logic**.
- Enhanced **cross-functional team productivity**, measured by faster blocker resolution and clearer weekly priorities, by leading **weekly team sync-ups** with developers and project stakeholders.
- Reduced **response misclassification** and improved chatbot relevance, measured by stronger domain-specific intent matching, by designing and training **intent-response mappings** for user scenarios.

PROJECTS

Market Volatility Risk Assessment | *Python, ARIMA, GARCH*

[GitHub](#)

- Built a **market-risk forecasting pipeline**, measured by volatility analysis on a **\$1,000,000 simulated multi-asset portfolio**, by applying **Python, ARIMA(1,1,0), and GARCH(1,1)** models.
- Identified **macroeconomic-driven volatility spikes**, measured around **FOMC, CPI, and GDP** release periods, by modeling asset-return behavior and comparing event-window risk patterns.
- Validated **time-series model reliability**, measured by **ADF/KPSS, JB/Shapiro, and ACF/PACF** diagnostics, by testing stationarity, normality, and autocorrelation assumptions before interpreting portfolio risk.

GDP Growth Forecasting | *Python, Ridge&Lasso, PCA*

[GitHub](#)

- Built a **pseudo out-of-sample GDP forecasting system** in **Python** comparing **six models** (Naive Mean, Random Walk, Ridge, Lasso, and **PCA Diffusion Index**) across one- and four-quarter horizons using **20 macroeconomic time series** pulled from **FRED**.
- Implemented **shrinkage regression** and **Principal Component Analysis** with **expanding-window cross-validation** to extract **diffusion indices** from a large **macroeconomic panel**, reducing **overfitting** on **quarterly time series data**.
- Evaluated **forecast accuracy** using **MSFE, MAE, and relative MSFE benchmarks**, finding that **big-data methods** outperformed the **random walk** at both horizons while the **naive mean** remained the strongest short-horizon benchmark.

Student Performance Analysis | *R, ggplot, tidyverse*

[GitHub](#)

- Identified **at-risk students**, measured by **GPA tiers**, by cleaning and grouping academic records in **R**.
- Improved **advisor insights**, measured by **major-grade patterns**, by creating **ggplot** visualizations.
- Supported **tutoring decisions**, measured by **performance trends**, by translating findings into recommendations.

Healthcare Prediction | *R, SQL, ggplot*

[GitHub](#)

- Analyzed **1,000+ healthcare records**, measured by **billing variation**, by using **SQL** and **R**.
- Tested **cost differences**, measured by **p-values and confidence intervals**, by applying hypothesis tests.
- Communicated **healthcare insights**, measured by **4+ visualizations**, by summarizing statistically valid results.

TECHNICAL SKILLS

Computer programming: Java, R, Python, SQL

Data Visualization: Tableau, PowerBI, Matplotlib

Tools & Software: STATA, SAS, Microsoft Office, GitHub

Extracurricular activities: Rutgers Consulting Group Member, Rutgers Quantitative Finance Club Member.